Spheroid For GJK

Let

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \tag{1.1}$$

$$\underline{r}=\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} a\cos\theta\sin\phi \\ b\sin\theta\sin\phi \\ c\cos\phi \end{bmatrix} \tag{1.2}$$

$$\underline{t}_{\phi}=\begin{bmatrix} a\cos\theta\cos\phi \\ b\sin\theta\cos\phi \\ -c\sin\phi \end{bmatrix} \tag{1.3}$$

$$\underline{t}_{\theta}=\begin{bmatrix} -a\sin\theta\sin\phi \\ b\cos\theta\sin\phi \\ 0 \end{bmatrix} \tag{1.4}$$

$$\begin{aligned}
\underline{n} &= \tilde{\underline{t}}_{\phi}\underline{t}_{\theta} \\
&= \begin{bmatrix} 0 & c\sin\phi & b\sin\theta\cos\phi \\ -c\sin\phi & 0 & -a\cos\theta\cos\phi \\ -b\sin\theta\cos\phi & a\cos\theta\cos\phi & 0 \end{bmatrix}\begin{bmatrix} -a\sin\theta\sin\phi \\ b\cos\theta\sin\phi \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} bc\cos\theta\sin^2\phi \\ ac\sin\theta\sin^2\phi \\ ab\cos\phi\sin\phi \end{bmatrix} \\
&= \begin{bmatrix} a\cos\theta\sin\phi \\ b\sin\theta\sin\phi \\ c\cos\phi \end{bmatrix}\frac{\sin\phi}{abc} \\
&= \underline{r}\frac{\sin\phi}{abc}
\end{aligned} \tag{1.5}$$

$$|\underline{n}|=|\underline{r}|\frac{|\sin\phi|}{abc} \tag{1.6}$$

$$|\underline{r}| = \sqrt{x^2 + y^2 + z^2} \tag{1.7}$$